

Lecture Note 6: Constructing Estimators

14.380: Statistical Methods in Economics, Fall 2026

1 Method of Analogy or Plug-In Principle

At the beginning of the course, we mentioned that we are interested in estimating parameters $\theta = T(F_{\mathcal{X}})$ that are functionals of the underlying cumulative distribution function of the data $\mathcal{X} = (X_1, \dots, X_n)$. Therefore, to construct an estimator, it seems natural to simply “plug in” an estimator of the cumulative distribution function $\hat{F}_{\mathcal{X}}$ into the functional

$$\hat{\theta} = T(\hat{F}_{\mathcal{X}}).$$

Moreover, given its desirable convergence properties, a natural choice would be to plug in the empirical distribution function. Indeed, we have already seen a number of such plug-in estimators. For example, $\mu = \mathbb{E}[X_i] = \int x dF(x)$ is a functional of the cdf. An analog estimator is $\hat{\mu} = \int x d\hat{F}(x) = \bar{X}$. If we wish to estimate $\theta = P\{X_i \in A\}$, we may use $\hat{\theta} = \frac{1}{n} \sum_{i=1}^n I\{X_i \in A\}$. Indeed, it can be shown formally that, provided the functional T is “well-behaved” (satisfies a particular differentiability condition), this plug-in estimator is consistent and asymptotically normal. Such results are introduced in other econometrics courses that cover empirical process theory.

Remark 1. The analogy principle presumes that the estimand can be written as a functional of the distribution of the *observable* data. For many estimands in economics this is immediate — means, variances, quantiles, regression coefficients. For causal estimands it is not: the average treatment effect is defined through potential outcomes, only one of which is ever observed per unit. Before the analogy principle can be applied, the researcher must supply an *identification* argument — assumptions under which the causal estimand equals some functional of the observable distribution. For example, if treatment is as-good-as-randomly assigned conditional on covariates X_i (“unconfoundedness”), then the average treatment effect satisfies

$$\tau = \mathbb{E}[\mathbb{E}[Y_i | X_i, D_i = 1] - \mathbb{E}[Y_i | X_i, D_i = 0]],$$

which is a functional of the joint distribution of (Y_i, D_i, X_i) — and the analogy principle then applies exactly as in this section. Identification arguments are a subject in their own right; the point here is that every estimator we construct in this lecture presupposes that this step has been settled.

2 Method of Moments

Let $X_1, \dots, X_n$ be a random sample from some distribution. Suppose that the k -dimensional parameter of interest θ satisfies the system of equations $\mathbb{E}[X_i] = m_1(\theta)$, $\mathbb{E}[X_i^2] = m_2(\theta)$, ..., $\mathbb{E}[X_i^k] = m_k(\theta)$ where $m_1, \dots, m_k$ are some known functions. Then the *method-of-moments estimator* $\hat{\theta}_{MM}$ of θ is the solution of the above system of equations when we substitute $\sum_{i=1}^n X_i/n$, $\sum_{i=1}^n X_i^2/n$, ...,

$\sum_{i=1}^n X_i^k/n$ for $\mathbb{E}[X_i]$, $\mathbb{E}[X_i^2]$, ..., $\mathbb{E}[X_i^k]$ correspondingly. In other words $\hat{\theta}_{MM}$ solves the following system of equations: $\sum_{i=1}^n X_i/n = m_1(\hat{\theta})$, $\sum_{i=1}^n X_i^2/n = m_2(\hat{\theta})$, ..., $\sum_{i=1}^n X_i^k/n = m_k(\hat{\theta})$. We implicitly assume here that the solution exists and is unique.

Example 1. Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$. Then $\mathbb{E}[X_i] = \mu$ and $\mathbb{E}[X_i^2] = \mu^2 + \sigma^2$. Thus, $\hat{\mu}_{MM} = \sum_{i=1}^n X_i/n$ and $\hat{\mu}_{MM}^2 + \hat{\sigma}_{MM}^2 = \sum_{i=1}^n X_i^2/n$. So $\hat{\sigma}_{MM}^2 = \sum_{i=1}^n X_i^2/n - (\sum_{i=1}^n X_i/n)^2$.

Example 2. Let $X_1, \dots, X_n$ be a random sample from an exponential distribution:

$$f(x; \lambda) = \lambda e^{-\lambda x} I\{x > 0\}.$$

One can calculate that $\mathbb{E}X_i = \frac{1}{\lambda}$. So one suggestion for an estimator is a solution to

$$\bar{X} = \frac{1}{\hat{\lambda}}$$

or

$$\hat{\lambda} = \frac{1}{\bar{X}}.$$

We may use higher moments as well. For example

$$\mathbb{E}X_i^2 = \frac{2}{\lambda^2}.$$

As such, another method-of-moments estimator for λ is:

$$\hat{\lambda} = \sqrt{\frac{2}{\frac{1}{n} \sum_{i=1}^n X_i^2}}$$

So, the method of moments estimator is not unique and depends on the moments chosen.

One can easily prove that if the function $m(\theta)$ has a unique inverse which is continuous, then the method-of-moments estimator is consistent (do this for your own practice! Hint: use the continuous mapping theorem). And if the inverse is continuously differentiable, we can use the delta method and prove asymptotic Gaussianity (try to do this as well).

The idea of the method-of-moments is a very old one. There is a generalization of it which allows for more moments than the dimensionality of the parameter and also allows for the data and parameter to be mixed up within the moment condition. It is called GMM (Generalized Method of Moments) and will be studied extensively later on, as the main workhorse of Econometrics.

Example 3. The method of moments turns economic assumptions into estimators. Suppose we are interested in the effect of schooling D_i on log earnings Y_i and posit

$$Y_i = \alpha + \beta D_i + U_i,$$

where U_i collects unobserved determinants of earnings. Least squares logic would impose $\text{Cov}(D_i, U_i) = 0$, but this is exactly what we doubt: individuals with higher unobserved earnings potential may acquire more schooling. Suppose instead that the researcher has an *instrument* Z_i — a variable that shifts schooling but is unrelated to the unobservables, such as a draft-lottery number or quarter of birth. The economic assumptions are the two moment conditions

$$\mathbb{E}[U_i] = 0 \quad \text{and} \quad \text{Cov}(Z_i, U_i) = 0.$$

Substituting $U_i = Y_i - \alpha - \beta D_i$, the second condition gives $\text{Cov}(Z_i, Y_i) = \beta \text{Cov}(Z_i, D_i)$, and so, provided $\text{Cov}(Z_i, D_i) \neq 0$ (the instrument is *relevant*),

$$\beta = \frac{\text{Cov}(Z_i, Y_i)}{\text{Cov}(Z_i, D_i)}.$$

The method of moments then replaces population covariances with sample covariances:

$$\hat{\beta}_{IV} = \frac{\widehat{\text{Cov}}(Z_i, Y_i)}{\widehat{\text{Cov}}(Z_i, D_i)}, \quad \hat{\alpha} = \bar{Y}_n - \hat{\beta}_{IV} \bar{D}_n.$$

This is the *instrumental variables estimator*. When Z_i is binary, $\hat{\beta}_{IV}$ reduces to a ratio of differences in means — the difference in average earnings between the two instrument groups divided by the difference in average schooling — so its sampling behavior can be analyzed with the tools we have already developed (the delta method, in particular). Notice the division of labor: economics supplies the moment conditions; statistics supplies the estimator and its sampling theory. Whether $\hat{\beta}_{IV}$ estimates a *causal* effect depends on the credibility of the moment conditions, which is an identification question in the sense of the remark above.

3 Maximum Likelihood Estimation

In this section we consider parametric estimation. We have a parametric estimation problem when we know the distribution of the data up to a finite-dimensional parameter θ (the only unknown part). We denote the joint pdf of $X = (X_1, \dots, X_n)$ as $f(x|\theta) = f(x_1, \dots, x_n|\theta)$ (for an i.i.d. sample we will have $f(x|\theta) = \prod_{i=1}^n f_1(x_i|\theta)$, where $f_1(x_i|\theta)$ is the pdf of one observation). That is, if we knew θ , we would know the exact distribution of the data.

Let $x = (x_1, \dots, x_n)$ denote the realization of $X = (X_1, \dots, X_n)$. By definition, the *maximum likelihood estimator* $\hat{\theta}_{MLE}$ of θ is the value that maximizes $f(x|\theta)$, i.e.

$$\hat{\theta}_{MLE} = \arg \max_{\theta \in \Theta} f(x_1, \dots, x_n|\theta).$$

The function $f(x|\theta)$, when considered as a function of θ for fixed values $x = (x_1, \dots, x_n)$, is called the *likelihood function*. It is usually denoted by $\mathcal{L}(\theta|x)$. Thus, the maximum likelihood estimator maximizes the likelihood function, which explains the name of this estimator. Since $\log(x)$ is

increasing in x , it is easy to see that $\hat{\theta}_{MLE}$ also maximizes

$$\ell(\theta|x_1, \dots, x_n) = \log \mathcal{L}(\theta|x_1, \dots, x_n).$$

Function $\ell(\theta|x_1, \dots, x_n)$ is called the *log-likelihood*. If $\ell(\theta|x_1, \dots, x_n)$ is differentiable in θ , then $\hat{\theta}_{MLE}$ satisfies the first-order condition (FOC): $\frac{d\ell}{d\theta}(\hat{\theta}_{MLE}|x_1, \dots, x_n) = 0$. If the data comes from an i.i.d. sample, then it is equivalent to $\sum_{i=1}^n \partial \log f_1(x_i|\hat{\theta}_{MLE})/\partial \theta = 0$. The reason we took the log of the likelihood function can now be seen: it is easier to take the derivative of the sum than the derivative of the product.

Finally, we call the function

$$S(\theta|x) = \partial \log f(x|\theta)/\partial \theta$$

the *score*. Thus, $\hat{\theta}_{MLE}$ solves the equation $S(\theta|x) = 0$.

Example 4. Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$. Then

$$\log f_1(x_i|\theta) = -\log \sqrt{2\pi} - (1/2) \log \sigma^2 - (x_i - \mu)^2/(2\sigma^2),$$

where $\theta = (\mu, \sigma^2)$. So

$$\ell(\theta|x_1, \dots, x_n) = -n \log \sqrt{2\pi} - (n/2) \log \sigma^2 - \sum_{i=1}^n (x_i - \mu)^2/(2\sigma^2)$$

The FOCs are

$$\partial \ell / \partial \mu = \sum_{i=1}^n (x_i - \mu) / \sigma^2 = 0,$$

$$\partial \ell / \partial \sigma^2 = -n/(2\sigma^2) + \sum_{i=1}^n (x_i - \mu)^2 / (2\sigma^4) = 0.$$

So $\hat{\mu}_{MLE} = \bar{X}_n$ and $\hat{\sigma}_{MLE}^2 = \sum_{i=1}^n (X_i - \bar{X}_n)^2 / n$.

Example 5. Let $X_1, \dots, X_n$ be a random sample from $U[0, \theta]$. Then $f_1(x_i|\theta) = 1/\theta$ if $x_i \in [0, \theta]$ and 0 otherwise. So $f(x_1, \dots, x_n|\theta) = 1/\theta^n$ if $0 \leq x_{(1)} \leq x_{(n)} \leq \theta$ and 0 otherwise. Thus, $\mathcal{L}(\theta|x) = (1/\theta^n)I\{\theta \geq x_{(n)}\}I\{x_{(1)} \geq 0\}$. We conclude that $\hat{\theta}_{MLE} = X_{(n)}$.

3.1 Fisher Information

Let $f(x|\theta)$ with $\theta \in \Theta$ be some parametric family. For given $\theta \in \Theta$, let $Supp_\theta = \{x : f(x|\theta) > 0\}$. $Supp_\theta$ is usually called the *support* of distribution $f(x|\theta)$. Assume that $Supp_\theta$ does not depend on θ .

As before, $\ell(\theta|x) = \log f(x|\theta)$ is called the log-likelihood function. Assume that $\ell(\theta|x)$ is twice continuously differentiable in θ for all $x \in Supp_\theta$ and $\left| \frac{\partial^2 \ell(\theta|x)}{\partial \theta^2} \right|$ is bounded by some function $g(x)$ such that $\mathbb{E}g(X) < \infty$ for random variable X with distribution $f(x|\theta)$. Then:

Definition 1. $I(\theta) = \mathbb{E}_\theta[(\partial \ell(\theta|X)/\partial \theta)^2]$ is called *Fisher information*.

Fisher information plays an important role in maximum likelihood estimation. The theorem below gives two information equalities:

Theorem 1. *In the setting above,*

$$(1) \mathbb{E}_\theta[\partial\ell(\theta|X)/\partial\theta] = 0$$

$$(2) I(\theta) = -\mathbb{E}_\theta[\partial^2\ell(\theta|X)/\partial\theta^2].$$

Proof. Since $\ell(\theta|x)$ is twice differentiable in θ , $f(x|\theta)$ is twice differentiable in θ as well. Let us differentiate the following identity with respect to θ :

$$\int f(x|\theta)dx \equiv 1.$$

The restrictions on dominance by $g(x)$ allow us to interchange differentiation and integration signs below:

$$\int \frac{\partial f(x|\theta)}{\partial\theta} dx = 0 \text{ for all } \theta \in \Theta.$$

The second differentiation with respect to θ yields

$$\int \frac{\partial^2 f(x|\theta)}{\partial\theta^2} dx = 0 \text{ for all } \theta \in \Theta. \quad (1)$$

Now notice that

$$\frac{\partial\ell(\theta|x)}{\partial\theta} = \frac{\partial \log f(x|\theta)}{\partial\theta} = \frac{1}{f(x|\theta)} \frac{\partial f(x|\theta)}{\partial\theta}$$

and

$$\frac{\partial^2\ell(\theta|x)}{\partial\theta^2} = -\frac{1}{f^2(x|\theta)} \left(\frac{\partial f(x|\theta)}{\partial\theta} \right)^2 + \frac{1}{f(x|\theta)} \frac{\partial^2 f(x|\theta)}{\partial\theta^2}.$$

The former equality yields

$$\mathbb{E}_\theta \left[\frac{\partial\ell(\theta|X)}{\partial\theta} \right] = \mathbb{E}_\theta \left[\frac{1}{f(X|\theta)} \frac{\partial f(X|\theta)}{\partial\theta} \right] = \int \frac{1}{f(x|\theta)} \frac{\partial f(x|\theta)}{\partial\theta} f(x|\theta) dx = \int \frac{\partial f(x|\theta)}{\partial\theta} dx = 0,$$

which is our first result. The latter equality yields

$$\mathbb{E}_\theta \left[\frac{\partial^2\ell(\theta|X)}{\partial\theta^2} \right] = - \int \frac{1}{f(x|\theta)} \left(\frac{\partial f(x|\theta)}{\partial\theta} \right)^2 dx,$$

here the second term disappears due to equation (1). So,

$$\begin{aligned} I(\theta) &= \mathbb{E}_\theta \left[\left(\frac{\partial\ell(\theta|X)}{\partial\theta} \right)^2 \right] = \int \left(\frac{1}{f(x|\theta)} \frac{\partial f(x|\theta)}{\partial\theta} \right)^2 f(x|\theta) dx \\ &= \int \frac{1}{f(x|\theta)} \left(\frac{\partial f(x|\theta)}{\partial\theta} \right)^2 dx = -\mathbb{E}_\theta \left[\frac{\partial^2\ell(\theta|X)}{\partial\theta^2} \right]. \end{aligned}$$

□

Example 6. Let us calculate Fisher information for one random draw from the $N(\mu, \sigma^2)$ distribution where σ^2 is known. Thus, our parameter $\theta = \mu$. The density of a normal distribution is $f(x|\mu) = \exp(-(x - \mu)^2/(2\sigma^2))/\sqrt{2\pi\sigma^2}$. The log-likelihood is $\ell(\mu|x) = -\log(2\pi\sigma^2)/2 - (x - \mu)^2/(2\sigma^2)$. So $\partial\ell(\mu|x)/\partial\mu = (x - \mu)/\sigma^2$ and $\partial^2\ell(\mu|x)/\partial\mu^2 = -1/\sigma^2$. So $-\mathbb{E}_\theta[\partial^2\ell(\mu|X)/\partial\mu^2] = 1/\sigma^2$. At the same time,

$$I(\theta) = \mathbb{E}_\mu[(\partial\ell(\mu|X)/\partial\mu)^2] = \mathbb{E}_\mu[(X - \mu)^2/\sigma^4] = 1/\sigma^2.$$

So, as was expected in view of the theorem above, $I(\theta) = -\mathbb{E}_\mu[\partial^2\ell(\mu|X)/\partial\mu^2]$ in this example.

Example 7. Let us calculate Fisher information for one random draw from a Bernoulli(θ) distribution. Note that a Bernoulli distribution is discrete. So we use a probability mass function (pmf) instead of a pdf. The pmf of Bernoulli(θ) is $f(x|\theta) = \theta^x(1 - \theta)^{1-x}$ for $x \in \{0, 1\}$. The log-likelihood is $\ell(\theta|x) = x \log \theta + (1 - x) \log(1 - \theta)$. So $\partial\ell(\theta|x)/\partial\theta = x/\theta - (1 - x)/(1 - \theta)$ and $\partial^2\ell(\theta|x)/\partial\theta^2 = -x/\theta^2 - (1 - x)/(1 - \theta)^2$. So

$$\begin{aligned} \mathbb{E}_\theta[(\partial\ell(\theta|X)/\partial\theta)^2] &= \mathbb{E}_\theta[(X/\theta - (1 - X)/(1 - \theta))^2] \\ &= \mathbb{E}_\theta[X^2/\theta^2] - 2\mathbb{E}_\theta[X(1 - X)/(\theta(1 - \theta))] + \mathbb{E}_\theta[(1 - X)^2/(1 - \theta)^2] \\ &= \mathbb{E}_\theta[X/\theta^2] + \mathbb{E}_\theta[(1 - X)/(1 - \theta)^2] \\ &= 1/(\theta(1 - \theta)), \end{aligned}$$

since $x = x^2$, $x(1 - x) = 0$, and $(1 - x) = (1 - x)^2$ if $x \in \{0, 1\}$. At the same time,

$$\begin{aligned} -\mathbb{E}_\theta[\partial^2\ell(\theta|X)/\partial\theta^2] &= \mathbb{E}_\theta[X/\theta^2 + (1 - X)/(1 - \theta)^2] \\ &= \theta/\theta^2 + (1 - \theta)/(1 - \theta)^2 \\ &= 1/\theta + 1/(1 - \theta) \\ &= 1/(\theta(1 - \theta)). \end{aligned}$$

So $I(\theta) = -\mathbb{E}_\theta[\partial^2\ell(\theta|X)/\partial\theta^2]$, as it should be.

Let us now consider Fisher information for a random sample. Let $X = (X_1, \dots, X_n)$ be an i.i.d. random sample from distribution $f_1(x_i|\theta)$. Then the joint pdf is $f(x|\theta) = \prod_{i=1}^n f_1(x_i|\theta)$ where $x = (x_1, \dots, x_n)$. The joint log-likelihood is $\ell_n(\theta|x) = \sum_{i=1}^n \ell_1(\theta|x_i)$. So Fisher information for the sample X is

$$I(\theta) = -\mathbb{E}_\theta \left[\frac{\partial^2 \ell_n(\theta|X)}{\partial\theta^2} \right] = -\mathbb{E}_\theta \sum_{i=1}^n \left[\frac{\partial^2 \ell_1(\theta|X_i)}{\partial\theta^2} \right] = nI_1(\theta).$$

Here $I_1(\theta)$ denotes Fisher information for one random draw from the distribution $f_1(x_i|\theta)$.

3.2 Rao-Cramer Bound

An important question in the theory of statistical estimation is whether there is a nontrivial bound such that no estimator can be more efficient than this bound. The theorem below is a result of this sort:

Theorem 2 (Rao-Cramer Bound). *Let $X = (X_1, \dots, X_n)$ be a random sample from distribution $f(x|\theta)$ with information $I(\theta)$. Let $W(X)$ be an estimator of θ such that*

$$(1) \frac{d}{d\theta} \mathbb{E}_\theta[W(X)] = \int W(x) \frac{\partial f(x|\theta)}{\partial \theta} dx, \text{ where } x = (x_1, \dots, x_n)$$

$$(2) \text{Var}(W) < \infty.$$

Then

$$\text{Var}(W) \geq \left(\frac{d}{d\theta} \mathbb{E}_\theta[W(X)] \right)^2 \frac{1}{I(\theta)}.$$

In particular, if W is unbiased for θ , then $\text{Var}(W) \geq \frac{1}{I(\theta)} = \frac{1}{nI_1(\theta)}$.

Proof. The first information equality gives $\mathbb{E}[S(\theta|X)] = \mathbb{E}_\theta \left[\frac{\partial \ell(\theta|X)}{\partial \theta} \right] = 0$. So,

$$\begin{aligned} \text{cov}(W(X), S(\theta|X)) &= \mathbb{E} \left[W(X) \frac{\partial \ell(\theta|X)}{\partial \theta} \right] \\ &= \int W(x) \frac{\partial \ell(\theta|x)}{\partial \theta} f(x|\theta) dx \\ &= \int W(x) \frac{\partial f(x|\theta)}{\partial \theta} \cdot \frac{1}{f(x|\theta)} f(x|\theta) dx \\ &= \int W(x) \frac{\partial f(x|\theta)}{\partial \theta} dx \\ &= \frac{d}{d\theta} \mathbb{E}_\theta[W(X)]. \end{aligned}$$

By the Cauchy-Schwarz inequality,

$$(\text{cov}(W(X), S(\theta|X)))^2 \leq \text{Var}(W(X)) \text{Var}(S(\theta|X)) = \text{Var}(W(X)) I(\theta).$$

Thus,

$$\text{Var}(W(X)) \geq \left(\frac{d}{d\theta} \mathbb{E}_\theta[W(X)] \right)^2 / I(\theta).$$

If W is unbiased for θ , then $\mathbb{E}_\theta[W(X)] = \theta$, $d\mathbb{E}_\theta[W(X)]/d\theta = 1$, and $\text{Var}(W(X)) \geq 1/I(\theta)$. $\square$

Example 8. Let us calculate the Rao-Cramer bound for random sample $X_1, \dots, X_n$ from a Bernoulli(θ) distribution. We have already seen that $I_1(\theta) = 1/(\theta(1-\theta))$ in this case. So Fisher information for the sample is $I(\theta) = n/(\theta(1-\theta))$. Thus, any unbiased estimator of θ , under some regularity conditions, has a variance no smaller than $\theta(1-\theta)/n$. On the other hand, let $\hat{\theta} = \bar{X}_n = \sum_{i=1}^n X_i/n$ be an estimator of θ . Then $\mathbb{E}_\theta[\hat{\theta}] = \theta$, i.e. $\hat{\theta}$ is unbiased, and $V(\hat{\theta}) = \theta(1-\theta)/n$ which coincides

with the Rao-Cramer bound. Thus, $\bar{X}_n$ is the uniformly minimum variance unbiased (UMVU) estimator of θ . The word “uniformly” in this situation means that $\bar{X}_n$ has the smallest variance among unbiased estimators for *all* $\theta \in \Theta$.

Example 9. Let us now consider a counterexample to the Rao-Cramer theorem. Let $X_1, \dots, X_n$ be a random sample from $U[0, \theta]$. Then $f_1(x_i|\theta) = 1/\theta$ if $x_i \in [0, \theta]$ and 0 otherwise. So $\ell_1(\theta|x_i) = -\log \theta$ if $x_i \in [0, \theta]$. Then $\partial \ell_1 / \partial \theta = -1/\theta$ and $\partial^2 \ell_1 / \partial \theta^2 = 1/\theta^2$. So $I_1(\theta) = 1/\theta^2$ while $-\mathbb{E}_\theta[\partial^2 \ell_1(\theta|X_i) / \partial \theta^2] = -1/\theta^2 \neq I_1(\theta)$. Thus, the second information equality does not hold in this example. The reason is that support of the distribution depends on θ in this example.

Moreover, consider an estimator $\hat{\theta} = ((n+1)/n)X_{(n)}$ of θ . Then $\mathbb{E}_\theta[X_{(n)}] = \frac{n}{n+1}\theta$, so that $\mathbb{E}_\theta[\hat{\theta}] = \theta$, and

$$V(\hat{\theta}) = ((n+1)^2/n^2)V(X_{(n)}) = \theta^2/(n(n+2))$$

as we saw when we considered order statistics. So $\hat{\theta}$ is unbiased, but its variance is smaller than $1/I_n(\theta) = \theta^2/n$, where $I_n(\theta) = nI_1(\theta) = n/\theta^2$ would be the Fisher information for the sample. Thus, the Rao-Cramer theorem does not work in this example either. Again, the reason is that the Rao-Cramer theorem assumes that support is independent of parameter.

Remark 2. The Rao-Cramér bound answers the question “how precisely can θ be estimated?” *within a parametric family*: it presumes the researcher knows the distribution of the data up to a finite-dimensional parameter. In modern causal inference, estimands such as the ATE are defined without any parametric model for the data. Remarkably, the question still has an answer: there exist *semiparametric efficiency bounds* that play exactly the role of the Rao-Cramér bound when the “parameter” is a functional like the average treatment effect and the model is left nonparametric. Hahn (1998) derives this bound for the average treatment effect under unconfoundedness; it depends on the propensity score and the conditional variances of the outcomes.